

2082:

Long Answer Questions: [10 marks each]

1. Describe the Feistel Cipher structure. Given the key {2B, 7E, 15, 16, 28, AE, D2, A6, AB, F7, 97, 66, 01, 02, 03, 04}, compute the first 4 byte of next key after first iteration, using the following S-Box in AES.

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76

2. Illustrate the man in middle attack in Diffie – Hellman key exchange protocol. Assume the prime number be 19 and 10 as its primitive root. Select 5 as private key and 4 as random integer. Find the cipher text of $M = 2$ using Elgamal crypto system.
3. List the properties of hash function. Describe the algorithm for SHA-1.

Short Answer Questions: [5 marks each]

4. Describe any three types of malicious logic.
5. Given the key “HELLOWORLD”, encrypt the plaintext “TURINGTEST” using Play fair cipher.
6. Explain any two modes of block cipher encryption.
7. State Fermat’s theorem with example. What is the implication of discrete logarithm?

8. Describe the working mechanism of digital signature algorithm.
9. List and explain the types of firewalls.
10. Decrypt the ciphertext “HI” using Hill Cipher where the key is $\begin{bmatrix} 5 & 4 \\ 3 & 3 \end{bmatrix}$.
11. What is digital certificate? Discuss the certificate life cycle.
12. Define authentication system. Discuss about challenge response system.

2081:

Long Answer Questions: [10 marks each]

1. Let us consider the 4 bits key set as {1100, 1010, 0000, 1111, 0101, 1001} and input text as {1011, 1110, 1011, 1000}. Now trace the first full round operation of IDEA algorithm.
2. What is Message Authentication Code? List the operation of computing digest value in different passes of MD4. Describe about Needham-Schroeder protocol. Does greedy algorithm guarantee optimal solution? Solve the Fractional knapsack problem to find maximum loot from given information.
3. Why do we need discrete logarithm? Illustrate with an example. Consider a Diffie-Hellman scheme with a common prime $p = 13$ between user A and user B. Suppose public key of A is 10 and public key of B is 8. Now determine their private keys and shared secret keys. Select any valid primitive root of 13.

Short Answer Questions: [5 marks each]

4. Show the encryption of plain text "ALGORITHM" using the key "PSEUDOCODE" using Playfair cipher.
5. Discuss the working mechanism of Kerberos protocol.
6. What is the use of firewall? How does circuit level gateway differ from stateful inspection firewall?
7. What is intrusion? Explain any two types of intrusion detection system.

8. Find the multiplicative inverse of polynomial $\{95\}$ using extended Euclidean Algorithm.
9. What is DoS attack? Discuss about PKI trust model.
10. Using Vigenère cipher with key = “worlds”, encrypt the plain text “hello everyone”.
11. Describe the different modes of block cipher.
12. Write short notes on (any two)
 - a. Totient value of any positive integer.
 - b. Properties of hash function.
 - c. Virus or Worms.

2080:

Long Answer Questions: [10 marks each]

1. Define discrete logarithms. How key generation, encryption and decryption are done in RSA. In an RSA cryptosystem, given $p=13$ and $q=7$, determine private key, public key and perform encryption and decryption for the text $M="hi"$ using 0 to 25 for letters from a to z.
2. Write down the encryption and decryption process at 2-DES and 3-DES. Explain the Feistel cipher structure. Divide $5x^2 + 4x + 6$ by $2x + 1$ over $GF(7)$.
3. What are the applications of hash functions? Discuss how SHA-1 algorithm generates hash value from a given message.

Short Answer Questions: [5 marks each]

4. Show encryption and decryption of "csit" using hill cipher having key, $k = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$.
5. Give the formal definition of authentication system. Describe about one way and mutual authentication system.
6. List the stage of certificate life cycle. What are the types of firewalls?
7. What is malicious logic? How are zombies different from trojan horses?
8. How is Miller Rabin test used for primality testing? Show whether the number 561 passes the test.

9. Show that the set of integers is Ring under addition and multiplication.
10. How are substitution ciphers different from transposition ciphers? Given a message
M=" CSIT PROGRAM IS A HOT CAKE", encrypt M using Rail Fence cipher with rail size 3.
11. Describe about IPSec. List the five services of PGP.
12. How does meet in middle attack work in Diffie Helman key exchange protocol? Explain.

Model:

Long Answer Questions: [10 marks each]

1. Define the CIA Triad and explain its components. List out the services of security.

Differentiate between block ciphers and stream ciphers with examples.
2. Given $p = 61$, $q = 53$. Calculate the public key, private key, and encrypt the message "42". Then, decrypt the ciphertext to retrieve the original message.
3. Explain the single round operation in DES. Describe the F function in DES.

Short Answer Questions: [5 marks each]

4. Explain the process of generating message digests using MD4.
5. Encrypt the message "INFORMATION" using the Playfair cipher with the keyword "SECURITY".
6. Explain the concept of digital signatures and differentiate between direct and arbitrated digital signatures.
7. Define firewall and types of firewalls.
8. State Fermat's theorem example. Write the algorithm for it.
9. Explain IP Security (IPSec) and its components.
10. Divide $3x^2 + x + 6$ by $5x + 3$ over GF (9).
11. Explain the process of public key distribution and secret key distribution using public key cryptography.

12. Explain the concept of a denial-of-service attack and provide examples.

2079:

Long Answer Questions: [10 marks each]

1. Illustrate the concept of security policy and mechanism with an example. Differentiate between block cipher and stream cipher. Explain the process of key expansion in AES.
2. Describe the properties of hash functions. Discuss how hash value is generated using SHA-1 algorithm.
3. Show that Z_5 is a field. John publishes the ElGamal public key $(q, \alpha, Y_A) = (101, 2, 14)$. Jane desired to send the secret message CSIT to John. Using the equivalence $A = 0, B=1, \dots, Z=25$, encrypt the message using John's public key. Use a random number $k = 4$.

Short Answer Questions: [5 marks each]

4. Differentiate between Trojan horse and virus. Describe any two types of intruders.
5. The message "IMOGUN" was encrypted with a Playfair cipher using keyword "GALOIS". Decrypt the message.
6. How encryption is done using IDEA algorithm.
7. Describe the services provided by Pretty Good Privacy protocol to secure email.
8. Define challenge response system. Why do we need Kerberos?
9. How direct digital signature is different from arbitrated digital signature? How digital signature generation and verification is done using RSA.
10. Why do we need discrete logarithm over normal logarithm? Find out whether 3 is primitive root of 7 or not.

11. Which one is more secure, monoalphabetic cipher or poly alphabetic cipher? Justify.

Using rail fence cipher encrypt the text "LEARNING AND TEACHING ARE DIFFERENT"
using 3 as rails.

12. What is the condition of for two integers, x and y , to be relatively prime? Find whether 61 is prime or not using Miller-Rabin algorithm.

2078:

Long Answer Questions: [10 marks each]

1. Define CIA triad. State the encryption process of double and triple DES. What is the task of S-Box in DES? Discuss with an example.
2. Explain the generic model of digital signature process. Consider the two prime numbers 7 and 19. Select 29 as public key and 41 as private key. Encrypt the plaintext 4 and decrypt the cipher text 3 using RSA.
3. Define Galois field with an example. Explain any two modes of block cipher encryption. Determine the quadratic residues of 7.

Short Answer Questions: [5 marks each]

4. What does intrusion mean? How does the system detect intrusion? List any four types of firewalls.
5. Decrypt the message "GVPI" using Hill cipher taking the key as {Row1 = 3 7, Row2 = 5 12}.
6. Describe the PKI trust model.
7. Define authentication system. Illustrate the need of mutual authentication over one way authentication with an example.
8. Find the value of $7^{2019} \text{ MOD } 13$ using Fermat's Little theorem. Define Euler totient function with an example.
9. List the properties of hash function. Discuss the first pass of MD4.

10. Differentiate between Symmetric and Asymmetric cipher. Encrypt the message “HELL” using the key “FAIL”.

11. Divide $3x^2 + 4x + 3$ by $5x + 6$ over GF (7).

12. Define SSL protocol. Mention the services provided by PGP.

2076:

Long Answer Questions: [10 marks each]

1. Among monoalphabetic and polyalphabetic cipher, which one is more vulnerable?
Justify your statement. Which types of keys are considered weak keys in DES? Explain the round operation in IDEA.
2. State Fermat's theorem with an example. Given the prime number $p=29$ and its primitive root $g=8$, private key sender with $X=9$ and random integer $K=11$, encrypt the message $m=13$ using ElGamal cryptosystem.
3. Compare the SHA parameters between SHA-1 and SHA-2 families. Decrypt the cipher text DRJI with the key (7 8, 11 11) using the Hill cipher.

Short Answer Questions: [5 marks each]

4. Define discrete logarithm. Explain the procedure of sharing the secret key in Diffie Hellman.
5. Distinguish between stream cipher and block cipher. Encrypt the message WE ARE IN SAME RACE UNTILL OVER LIVE END using Rail fence cipher using 4 as a number of rails.
6. Define digital signature. Describe the approaches of DSS.
7. What is the task of a firewall? List the elements of X.509.
8. How does the nature of worms differ from viruses? Define PKI with its architecture model.
9. Explain the procedure of mix column transformation in AES with an example.

10. What is the role of the prime number in the Euler totient Function? Find the GCD of 12 and 16 using the Euclidean algorithm.

11. Write down any two limitations of MAC. What do policy and mechanism mean in cryptography? Describe with a scenario.

12. Write short notes on:

- a. Classes of Intruder
- b. SSL
- c. DoS Attack